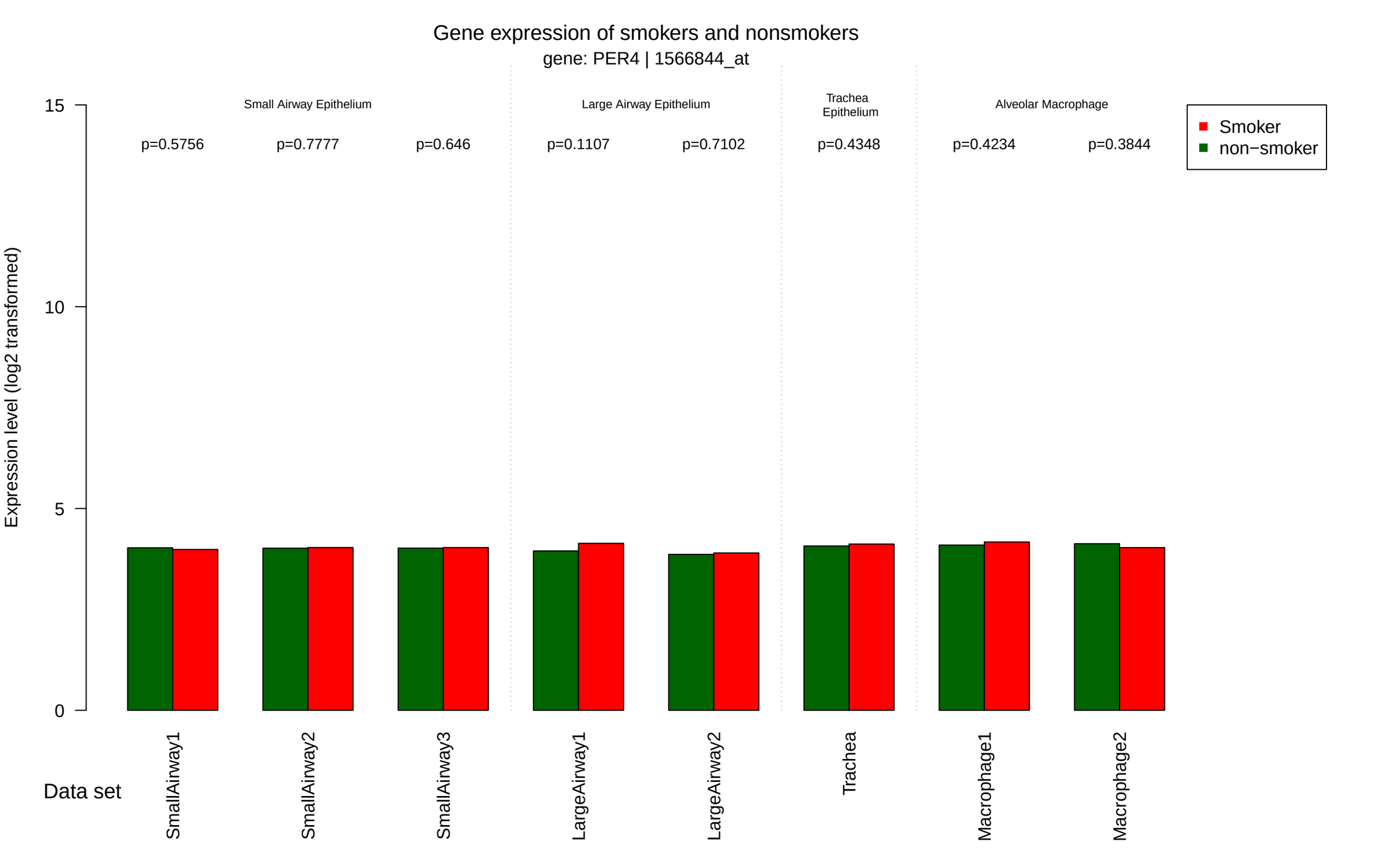
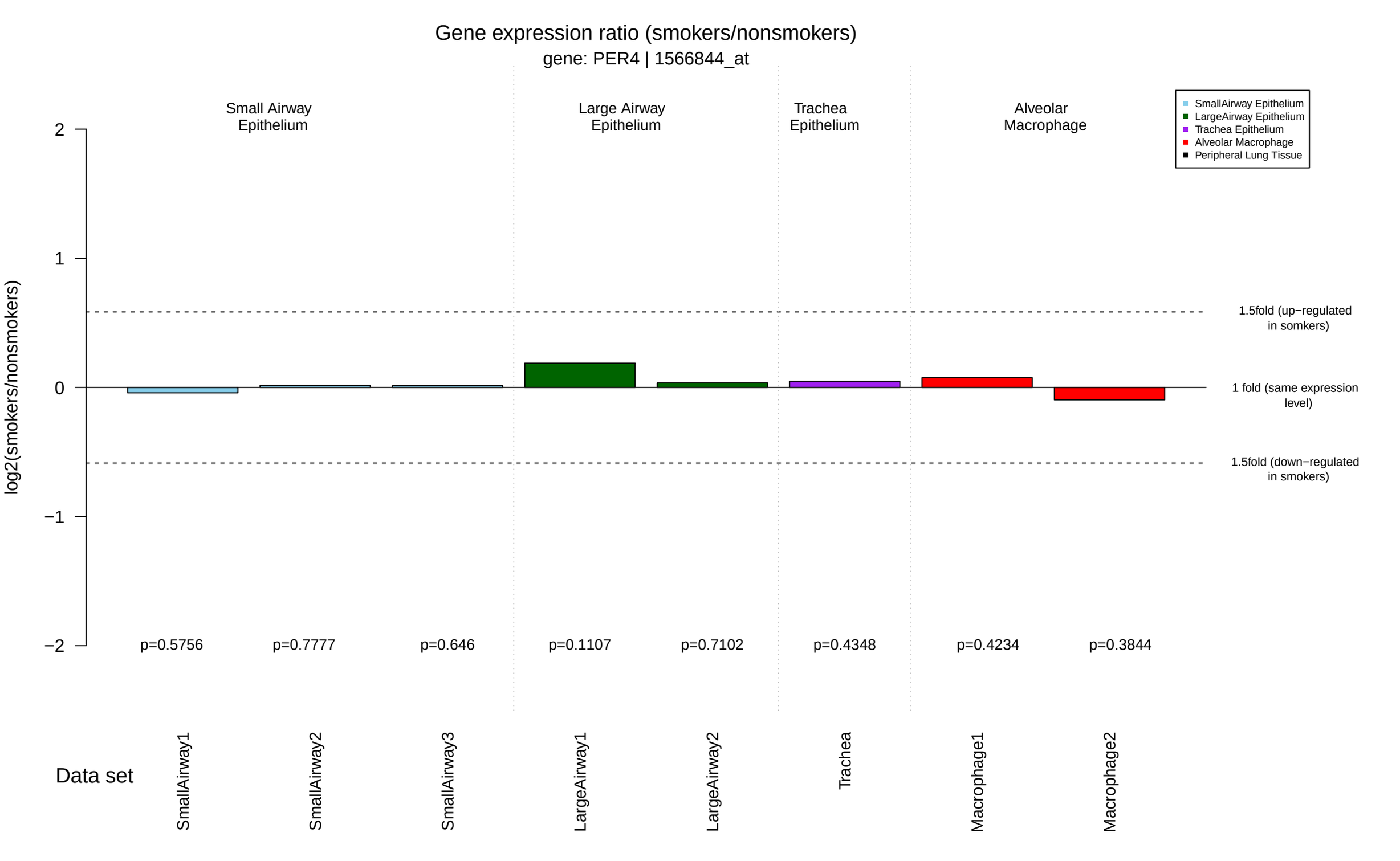
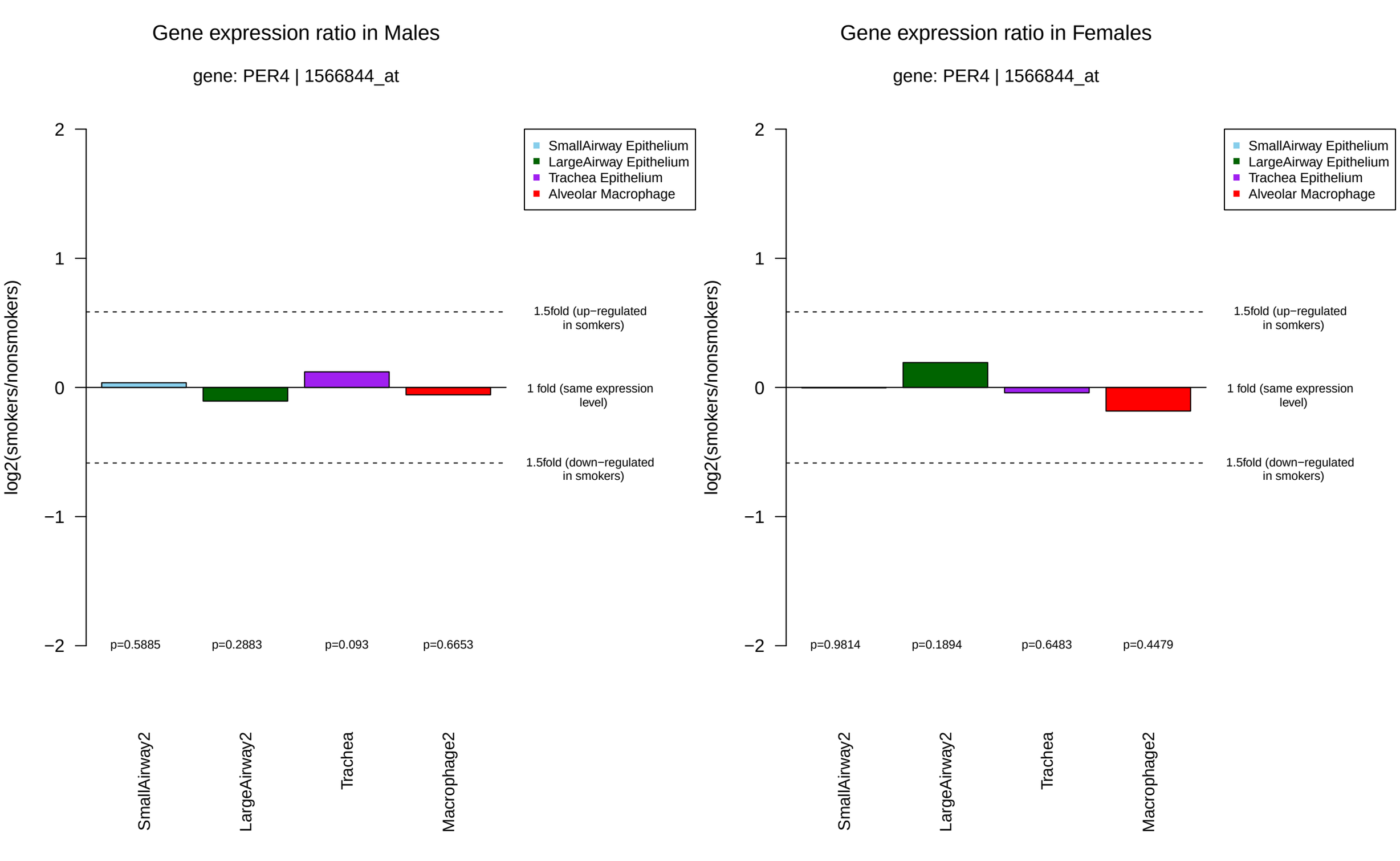
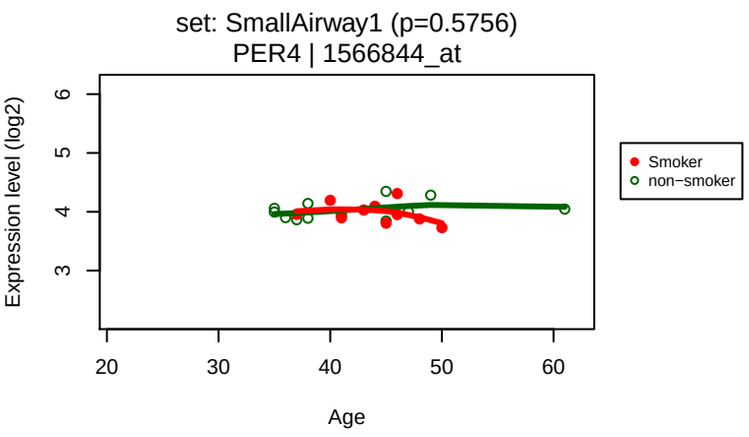
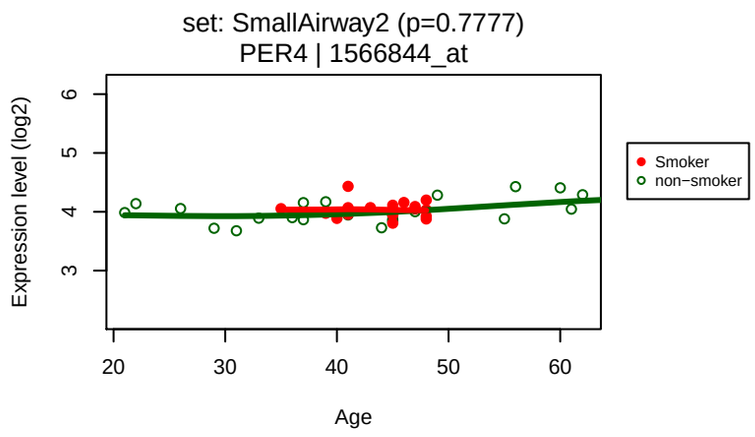
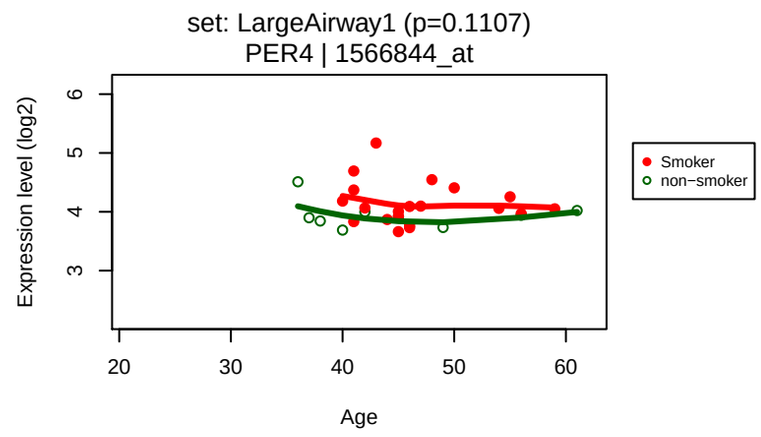

Gene expression of smokers and nonsmokers
gene: PER4 | 1566844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5756
p=0.7777
p=0.646
p=0.1107
p=0.7102
p=0.4348
p=0.4234
p=0.3844
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

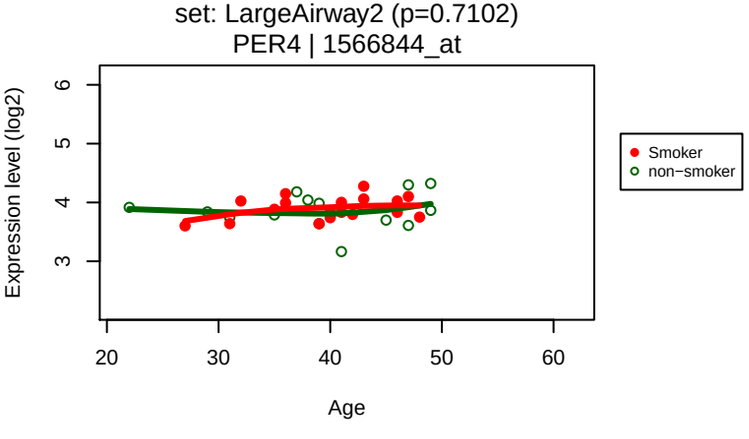

Gene expression ratio (smokers/nonsmokers)
gene: PER4 | 1566844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5756
p=0.7777
p=0.646
p=0.1107
p=0.7102
p=0.4348
p=0.4234
p=0.3844
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

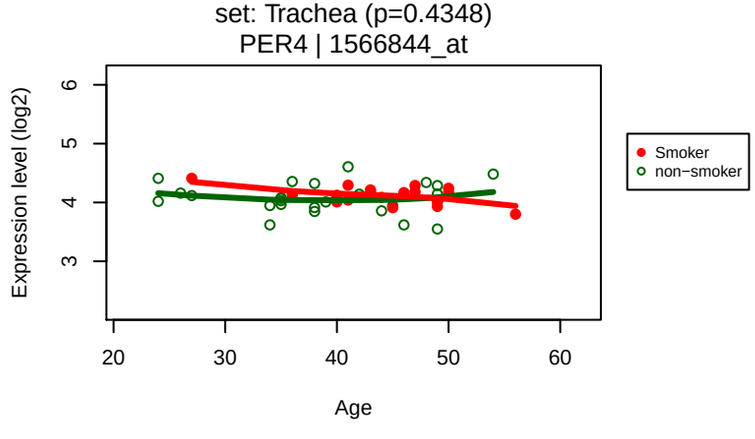

Gene expression ratio in Males
gene: PER4 | 1566844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5885
p=0.2883
p=0.093
p=0.6653
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER4 | 1566844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9814
p=0.1894
p=0.6483
p=0.4479
SmallAirway2
LargeAirway2
Trachea
Macrophage2

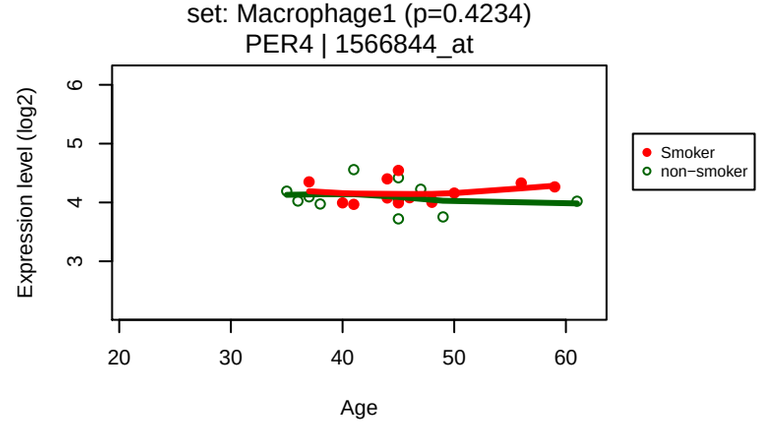

set: SmallAirway1 (p=0.5756)
PER4 | 1566844_at
Expression level (log2)
Age
Smoker
non−smoker

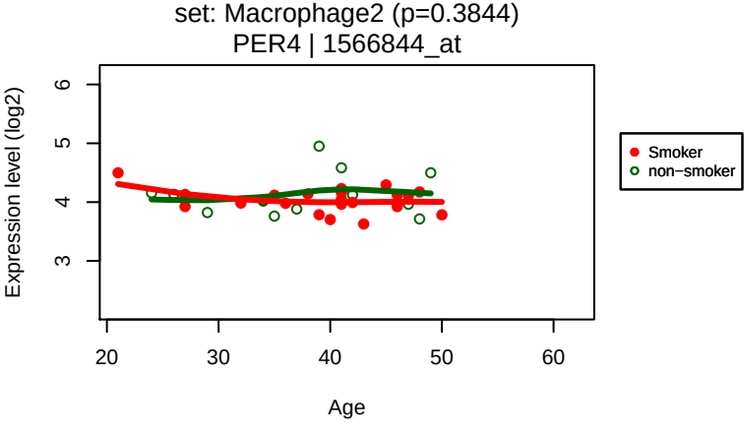

set: SmallAirway2 (p=0.7777)
PER4 | 1566844_at
Expression level (log2)
Age
Smoker
non−smoker

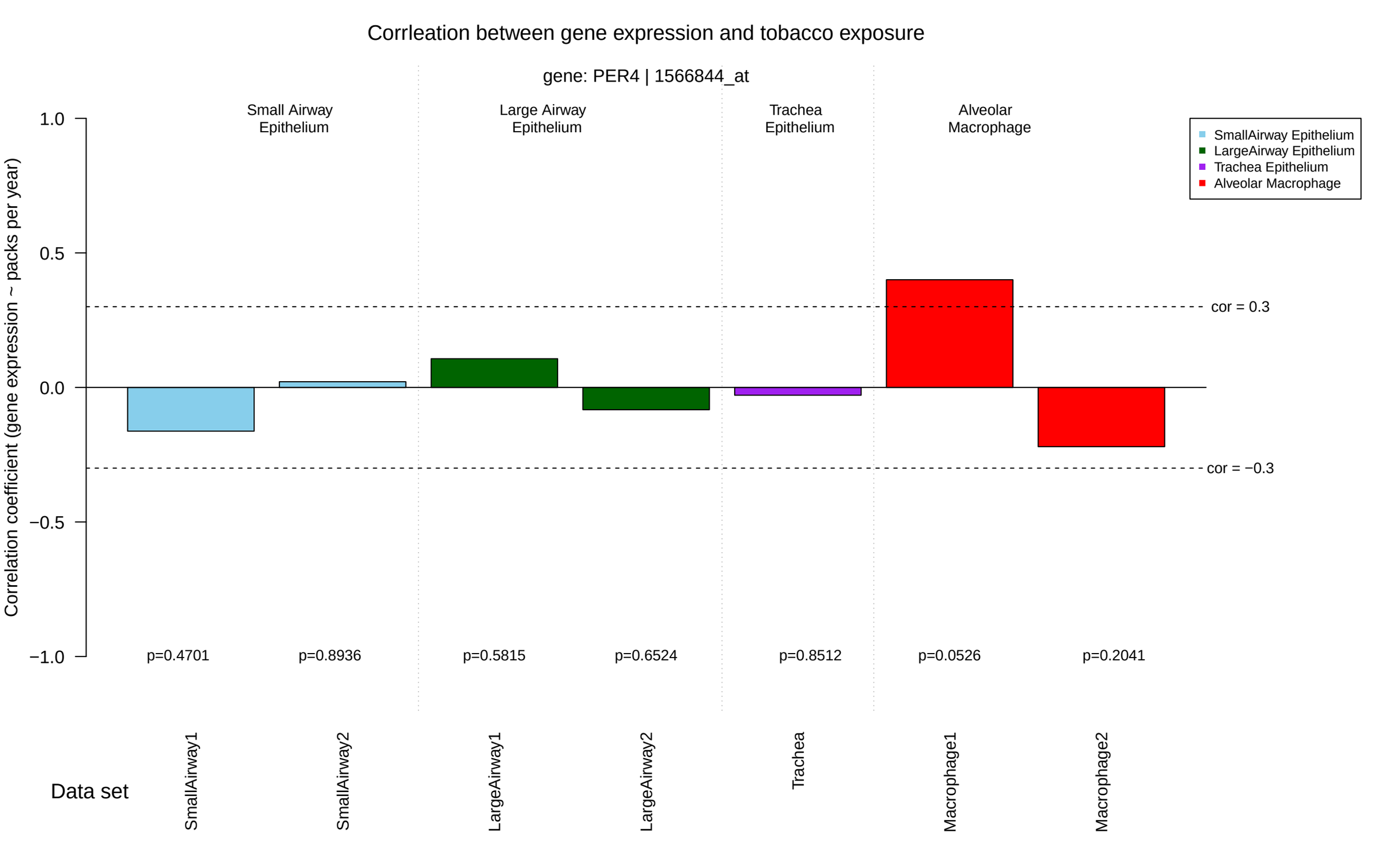

set: LargeAirway1 (p=0.1107)
PER4 | 1566844_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7102)
PER4 | 1566844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4348)
PER4 | 1566844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4234)
PER4 | 1566844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3844)
PER4 | 1566844_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER4 | 1566844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4701
p=0.8936
p=0.5815
p=0.6524
p=0.8512
p=0.0526
p=0.2041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2